

Stage 2.5 — Final Rules Validation

1. Remaining Issues Found

Issue 1 — Anita's ₹6,000 EMI was presented as if it were a fact *Decision:* Strike ₹6,000. Replace with the canonical fallback value derived from Issue 2's final rule (below), explicitly labeled as a demonstration assumption, not a Lokta-provided fact. *Final rule:* Persona walkthroughs must tag every number as **FACT** (from Lokta's text), **USER ANSWER** (hypothetical, as a real user would type it), **ASSUMPTION** (only because Lokta's text doesn't supply it), or **DERIVED** (computed). Anita's monthly high-cost-debt payment is an **ASSUMPTION**, computed via the same fallback formula the live app would use — never an invented standalone number. *Reason:* A demonstration must exercise the actual product rule, not a number picked for narrative convenience — otherwise the walkthrough isn't actually validating the rules engine.

Issue 2 — Unknown high-cost-debt EMI fallback was arbitrary (flat 20%) *Decision:* Hybrid of Option C (ask first) + a documented conservative-range fallback, not left unresolved. *Final rule:*

- The question is asked (already a required adaptive question, Stage 2) whenever `high_cost_debt` amount > 0.
- **If answered:** use the stated figure directly. Confidence: not penalized on this basis alone.
- **If unknown:** compute a plausible range — low = outstanding × 15% (a ~7-month informal amortization), high = outstanding × 30% (a ~3–4-month amortization, closer to how many short-tenure app/informal loans actually structure repayment in India). **The single point value used in all downstream formulas is outstanding × 25%** — the conservative-leaning point within that range, chosen deliberately over the midpoint because understating this burden is the dangerous direction of error (it could produce a false "safe to borrow" signal).
- The range (15%–30%, i.e., ₹X–₹Y) is shown to the borrower alongside the single figure actually used, so the estimate is never presented as more precise than it is.
- **Confidence effect:** any output depending on a *defaulted* (non-user-provided) high-cost-debt EMI is capped at LOW. *Category:* Product assumption / My judgement.

Issue 3 — Unknown household expenses (esp. Ravi) *Decision:* Agree with the stated preference, implemented pragmatically for a 12–16h build — **not** a full negotiation loop, but a single cheap clarifying step before falling back to a default. *Final rule:*

1. `household_expenses` remains a **must** question, estimate allowed.

2. **If the borrower gives no estimate at all**, the app asks ONE lightweight coarse-bucket follow-up: "*Roughly, are your monthly essentials closer to ₹15,000 / ₹25,000 / ₹40,000 / more?*" If answered, use that bucket's representative value as the point estimate, confidence **MEDIUM** for affordability outputs (not full precision, but not a blind default either).
3. **If still refused/unanswered**: use the default (55%/65% of income by segment), but the **safe amount is shown as a range**, computed at both a low-expense (default – 15 pts) and high-expense (default + 15 pts) scenario, not a single precise figure. Confidence **LOW**.
4. **Expenses > income**: disposable floors at ₹0 immediately; flagged as a pre-existing affordability concern independent of the new loan (unchanged from Stage 2 edge-case handling). *Reason*: one cheap follow-up materially improves precision at near-zero engineering cost; a full "ask until answered" loop isn't worth building in this timebox, and showing a range when truly unanswered is honest rather than falsely precise.

Issue 4 — productive_return Decision: Option B — retained, strictly qualitative. *Final rule*: the field may only affect the **explanation text and Negotiation Card framing** ("this is a productive purchase — worth reassessing if it increases income as expected"). It is hard-coded to never touch lenderCapacity, safeCapacity, or fairRate. This guardrail is enforced structurally in Stage 3 by keeping the field entirely outside the calculation-layer function signatures — the calculation functions never receive it as an argument. *Reason*: keeping it costs nothing and adds borrower-relevant color; the risk (self-reported ROI inflating hard numbers) is eliminated by scope, not by discipline alone.

Issue 5 — Fair rate needed a deterministic algorithm Decision: adopt a position-on-a-0–100-scale model, one named adjustment per independent factor, no factor scored twice. Full algorithm in Section 6 below.

Issue 6 — Every persona recalculated from scratch Done in Sections 13–15, with every input tagged FACT / USER ANSWER / ASSUMPTION / DERIVED.

2. Numerical Rule Audit

Rule	Final value	Why	Source/Judgement	Used where
FOIR — salaried	50%	Standard unsecured affordability heuristic	My judgement	Lender capacity

Rule	Final value	Why	Source/Judgement	Used where
FOIR — self-employed/ITR	45%	Self-certified income, more caution	My judgement	Lender capacity
FOIR — self-employed/informal undocumented	35%	Reflects real underfinancing of cash income	My judgement	Lender capacity
FOIR — secured	60%	Collateral reduces lender loss-given-default	My judgement	Lender capacity
LTV — gold	85%/80%/75% tiered by ticket size	RBI's 2026 tiered LTV rule	External fact	Lender capacity (gold)
LTV — LAP residential	70%	Market rate-card research	External fact	Lender capacity
LTV — LAP commercial	60%	iServeFinancial market comparison	External fact (range 60–65%, conservative end chosen)	Ravi's lender capacity
Collateral valuation haircut	20%	Self-reported values are unverified/optimistic	My judgement	Lender capacity (secured)
Retention factor — salaried stable	50%	Below lender FOIR by design	My judgement	Safe capacity
Retention factor — salaried less stable	40%	"	My judgement	Safe capacity
Retention factor — self-	40%	"	My judgement	Safe capacity

Rule	Final value	Why	Source/Judgement	Used where
employed/ITR steady				
Retention factor — self-employed seasonal/undoc	30%	"	My judgement	Safe capacity
Retention factor — informal/gig	25%	Highest volatility	My judgement	Safe capacity
Retention floor / cap	10% / 55%	Prevents absurd extremes	My judgement	Safe capacity
Variable-income adjustment	-5 pts	Excludes unreliable slice	My judgement	Safe capacity
Savings <1 month adjustment	-5 pts	No buffer	My judgement	Safe capacity
Savings >6 months adjustment	+5 pts	Demonstrated self-insurance	My judgement	Safe capacity
Dependents adjustment	-5 pts (>2 dependents, no other earner)	Larger single-income households have less slack	My judgement	Safe capacity
Recent-bounce adjustment (safe capacity)	-10 pts	Current obligations already stretched	My judgement	Safe capacity
High-cost-debt soft adjustment (safe capacity)	-10 pts	Distinct from bounce; separate signal	My judgement	Safe capacity
Upcoming-expense adjustment	-5 pts	Foreseeable shock should be pre-funded	My judgement	Safe capacity

Rule	Final value	Why	Source/Judgement	Used where
High-cost debt definition	APR $\geq$ 30%	Recognized predatory-adjacent threshold	My judgement	High-cost-debt rule (single definition, used everywhere)
High-cost debt SEVERE hard stop	burden $\geq$ 30% of eligible safe income	Burden alone is disqualifying	My judgement	Decision tree
High-cost debt COMPOUND hard stop	burden $\geq$ 15% AND recent bounce	Two independent stress signals together	My judgement	Decision tree
High-cost debt SOFT signal	burden < 15%, no bounce	Insufficient alone	My judgement	Decision tree (soft-signal escalation only)
Unknown high-cost-debt EMI fallback	outstanding $\times$ 25% (range shown 15–30%)	Conservative-leaning point estimate	My judgement	Safe capacity, Decision, Stress
Income shock magnitude	–20%	Brief's own suggested example	Assignment-derived (adopted default)	Stress test
Rate shock magnitude	+2 points	Brief's own suggested example	Assignment-derived (adopted default)	Stress test
Stress classification bands	$\leq$ 35 Comfortable / 36–45 Tight / 46–55 Stressed / >55 Unsustainable	Bracket common FOIR ceilings, one clean set	My judgement	Stress test

Rule	Final value	Why	Source/Judgement	Used where
Fair-rate starting position	50 (of 0–100)	Midpoint = no information advantage either way	My judgement	Fair rate
Fair-rate base half-width	15 position-points	Non-trivial band even with full information	My judgement	Fair rate
Fair-rate per-unknown widening	+10 pts/side per unknown, capped so total half-width ≤ 50	Confidence should visibly cost band width	My judgement	Fair rate
Credit score adjustment	–20 (≥ 750) / –10 (700–749) / 0 (650–699) / +15 (550–649) / +30 (< 550) / +5 (thin-file) / 0 (unknown, +widening)	Standard tiering; thin-file $\neq$ bad score	My judgement	Fair rate
Repayment history adjustment	–5 (clean) / +20 (recent bounce) / 0 (unknown, +widening)	Bounce is a strong, singular signal	My judgement	Fair rate
Stability adjustment	–10 (stable) / –5 (moderate) / +10 (unstable) / 0 (unknown, +widening)	"	My judgement	Fair rate
Documentation adjustment	–5 (full) / 0 (partial) / +15 (none) / +5 (unknown status)	"	My judgement	Fair rate

Rule	Final value	Why	Source/Judgement	Used where
High-cost-debt-present adjustment (rate)	+10	Distinct from repayment-history factor — no double count	My judgement	Fair rate
Presentation Headroom Factor	90% (Recommended EMI = Safe EMI × 0.90)	Universal margin for self-report error, applied once, never purpose-based	My judgement	Recommended amount only
Request/safe ratio soft-signal trigger	>1.5×	Round, explainable	My judgement	Decision tree
Confidence bands	HIGH (0 missing material inputs) / MEDIUM (1–2) / LOW (3+ or any heavily-weighted input missing)	Countable, defensible rule over an invented weighted score	My judgement	All outputs
Household-expense default	55% (salaried) / 65% (self-employed/informal) of income; ±15 pts range if fully defaulted	Placeholder only, always disclosed	My judgement	Safe capacity
Existing-EMI unknown floor	10% of income	Avoids overstating capacity from an unknown	My judgement	Lender/safe capacity
Tenure defaults	Personal 36 / Gold 12 / LAP 84 / Business 36 / Two-wheeler 36 / Home 180 (months)	Aggregator/market norms	My judgement	EMI conversion

Rule	Final value	Why	Source/Judgement	Used where
Processing fee assumption	1–2.5% of principal	Aggregator-observed typical range	External fact (approximate)	Cost model
Income-range-to-point rule	Midpoint if self-employed, tenure ≥3 yrs; lower bound if gig/daily-wage	Distinguishes normal seasonal variation from acute volatility	My judgement	Income model
Documentation haircut — lender, unsecured	10% of undocumented portion counted	Lenders can't verify unsecured cash income	My judgement	Lender capacity
Documentation haircut — lender, secured	40% of undocumented portion counted	Character/bank-statement lending more common once collateral backs the loan	My judgement	Lender capacity
Documentation haircut — safe capacity	0% (full claimed income counted)	Provability to a bank is irrelevant to the borrower's real cash flow	My judgement	Safe capacity

No duplicate or conflicting values remain — every threshold above appears exactly once as the canonical value, cross-referenced by "used where," and the high-cost-debt tree in particular (Issue raised in Section 12 of the brief) now has exactly three states (severe / compound / soft) with no other threshold set anywhere else in the system.

3. Final Income Model

`documentedIncome` = ITR monthly-equivalent, or salary, or verified bank-statement average

`claimedTotalIncome` = self-employed (tenure ≥3y) range → MIDPOINT

gig/daily-wage range → LOWER BOUND

salaried / point figure → as given

$\text{undocumentedPortion} = \text{MAX}(0, \text{claimedTotalIncome} - \text{documentedIncome})$

$\text{eligibleIncomeLender} = \text{documentedIncome} + \text{haircutLender} \times \text{undocumentedPortion}$

$\text{haircutLender} = 10\% \text{ (unsecured)} / 40\% \text{ (secured)}$

$\text{eligibleIncomeSafe} = \text{claimedTotalIncome}$ (no documentation haircut)

Why lender and safe legitimately differ: a lender's affordability math is bounded by what it can verify; the borrower's own ability to make a payment is bounded by what they actually receive, verifiable or not. Both are true simultaneously, and the gap between them is real information, not a modeling shortcut.

Co-applicant income (Ravi's wife): added to *both* $\text{eligibleIncomeLender}$ and $\text{eligibleIncomeSafe}$ only when explicitly disclosed as a co-applicant, at its full documented value (her salary is fully verifiable) — never assumed, never partially counted.

4. Final Lender Capacity Model

$\text{maxTotalDebtService} = \text{eligibleIncomeLender} \times \text{FOIR}(\text{productType}, \text{incomeType}, \text{documentation})$

$\text{availableNewEMI} = \text{maxTotalDebtService} - \text{existingEMI} - \text{businessDebtEMI} - \text{highCostDebtEMI}$

$\text{FOIR-supportedAmount} = \text{EMI} \rightarrow \text{Principal}(\text{availableNewEMI}, \text{fairRateBandMidpoint}, \text{defaultTenure})$

Secured products only:

$\text{adjustedCollateralValue} = \text{statedCollateralValue} \times (1 - 0.20)$

$\text{LTV-supportedAmount} = \text{adjustedCollateralValue} \times \text{maxLTV}(\text{product}, \text{collateralType})$

$\text{Lender-likelyAmount} = \text{FOIR-supportedAmount}$ (unsecured)

$= \text{MIN}(\text{FOIR-supportedAmount}, \text{LTV-supportedAmount})$ (secured)

If collateral value is unknown: LTV-supportedAmount is not computed; secured lender-likely falls back to FOIR-only; confidence on that output floors at LOW; Card states the secured route is contingent on a formal valuation.

5. Final Safe Capacity Model

disposableCashFlow = eligibleIncomeSafe – essentialExpenses – existingEMI – businessDebtEMI

– highCostDebtEMI

adjustedRetentionFactor = baseRetentionFactor(profile)

+ variableIncomeAdj + savingsAdj + dependentsAdj

+ bounceAdj + highCostDebtSoftAdj + upcomingExpenseAdj

, clamped to [10%, 55%]

safeEMI (ceiling) = disposableCashFlow × adjustedRetentionFactor ← O4, the hard "do not

cross" line, never

discounted further

recommendedEMI = safeEMI × 0.90 ← applied exactly ONCE,

only to derive O2's

recommended amount —

never applied again

when computing the

ceiling itself

safeAmount (ceiling) = EMI→Principal(safeEMI, fairRateBandCeiling, defaultTenure)

recommendedAmount = EMI→Principal(recommendedEMI, fairRateBandCeiling, defaultTenure)

If essentialExpenses is fully defaulted (Issue 3): compute safeAmount at both default – 15pts and default + 15pts and present as a range; otherwise present as the single-point-derived ceiling.

6. Final Fair Rate Algorithm (deterministic)

baseProductBand = [lowerRate, upperRate] (tier chosen: bank-tier if credit score ≥ 700 OR

strong collateral/tenure offsets a thin file;
NBFC-tier otherwise)

startingPosition = 50 (0–100 scale; 0 = lowerRate, 100 = upperRate)

adjustments (apply exactly once each, from the tables below):

- + creditScoreAdj
- + repaymentHistoryAdj
- + stabilityAdj
- + documentationAdj
- + highCostDebtPresentAdj (+10 if any high-cost debt outstanding, else 0)

finalPosition = clamp(startingPosition + Σ adjustments, 0, 100)

halfWidth = 15 + 10 × (count of unknowns among {creditScore, repaymentHistory, stability, documentation}), capped so finalPosition ± halfWidth stays within a total span ≤ 100

bandLowPosition = clamp(finalPosition – halfWidth, 0, 100)

bandHighPosition = clamp(finalPosition + halfWidth, 0, 100)

fairRateLower = lowerRate + (bandLowPosition/100) × (upperRate – lowerRate)

fairRateUpper = lowerRate + (bandHighPosition/100) × (upperRate – lowerRate)

Factor	Adjustment
Credit score ≥ 750	-20
Credit score 700–749	-10
Credit score 650–699	0
Credit score 550–649	+15
Credit score < 550	+30
Thin-file (never borrowed)	+5
Credit score unknown	0 (+ counts as one "unknown" for widening)
Repayment history clean	-5
Repayment history: recent missed/bounce	+20
Repayment history unknown	0 (+ widening)
Income/business stability: stable	-10
Stability: moderate	-5
Stability: unstable	+10
Stability unknown	0 (+ widening)
Documentation: full	-5
Documentation: partial (e.g. ITR below claimed cash)	0
Documentation: none	+15
Documentation status unknown	+5 (+ widening)
High-cost debt present	+10

This is deterministic, has exactly one slot per named factor (no factor is ever scored twice — e.g., a bounce affects repaymentHistoryAdj here and separately affects the decision tree and safe-capacity retention factor, but each of those is a **different output**, not a repeated hit on the same rate calculation), and produces identical results for any two engineers implementing it from this table.

7. Final Effective Cost Algorithm

$\text{netProceeds} = \text{Principal} - (\text{processingFee} + \text{otherDisclosedMandatoryFees})$

$\text{EMI} = \text{Principal} \times r \times (1+r)^n / ((1+r)^n - 1)$ [r = nominal monthly rate]

Solve numerically for r' (effective monthly rate) such that:

$\text{netProceeds} = \text{EMI} \times [1 - (1+r')^{-n}] / r'$ (bisection or Newton–Raphson in Stage 3)

Effective Annualized Borrowing Cost = $(1 + r')^{12} - 1$

Terminology: "Estimated Effective Annualized Borrowing Cost" — never called the lender's regulatory APR, since GST-on-fees, exact disbursal timing, and foreclosure terms are unknown. Shown as a range across the 1–2.5% fee assumption.

Priya, verified: recommended principal ₹6,55,000 (Section 13), nominal 11.5%, tenure 36 mo → EMI ₹21,600, fee 2% (₹13,100), net proceeds ₹6,41,900. Solving numerically: $r' \approx 1.078\%/month \rightarrow$ **annualized $\approx 13.7\%$** against an 11.5% nominal rate — the ~2.2-point gap is the fee's effect alone, confirmed by iteration ($r'=0.0108$ gives $RHS \approx ₹6,41,300$, $r'=0.01075$ gives $\approx ₹6,42,900$; interpolating to the ₹6,41,900 target lands at $r' \approx 0.01078$, annualized 13.7%).

8. Final Stress Algorithm

Unchanged from Stage 2, now explicitly noted as including highCostDebtEMI in the ratio every time:

$\text{stressedRatio}(\text{income scenario}) = (\text{recommendedEMI} + \text{existingEMI} + \text{businessDebtEMI} + \text{highCostDebtEMI})$

$/ (\text{income} \times 0.80)$

$\text{stressedRatio}(\text{rate scenario}) = \text{same numerator recomputed at } (\text{rate} + 2\text{pts}) / \text{unchanged income}$

classify: $\leq 35\%$ Comfortable | 36–45% Tight | 46–55% Stressed | $> 55\%$ Unsustainable

9. Final Decision Tree

STEP 1 — Hard stops (any ONE independent of amount comparison):

- a) $\text{highCostDebtBurden} \geq 30\%$ of $\text{eligibleIncomeSafe}$ (SEVERE)
 - b) $\text{highCostDebtBurden} \geq 15\%$ of $\text{eligibleIncomeSafe}$ AND $\text{recentBounce} = \text{true}$ (COMPOUND)
 - c) $\text{recentBounce} = \text{true}$ AND baseline (pre-shock) stress classification = Unsustainable
- any match: DON'T BORROW

STEP 2 — else, compare requested vs. safeAmount vs. $\text{lenderLikelyAmount}$:

- $\text{requested} \leq \text{safeAmount}$ → BORROW
- $\text{safeAmount} < \text{requested} \leq \text{lenderLikelyAmount}$ → BORROW LESS (recommend safeAmount)
- $\text{requested} > \text{lenderLikelyAmount}$ → BORROW LESS at lender ceiling, flagged as also above safe

STEP 3 — Soft signals (any TWO OR MORE escalate Step-2's verdict by one notch):

- $\text{emergencySavings} < 1$ month · recentBounce alone (not meeting 1c) · stress = Stressed
-
- $\text{requested} > 1.5 \times \text{safeAmount}$ · high-cost debt present but below hard-stop thresholds

10. Final Unknown-Data Rules

Unknown	Rule
Credit score	0 rate-position adjustment (neutral), +1 unknown for widening; never assumed 300
Household expenses	Coarse-bucket re-ask; if still unknown, default (55%/65%) shown as a range, confidence LOW
Existing EMI	Floor of 10% of income substituted, never ₹0
High-cost-debt EMI	outstanding × 25% (range 15–30% shown), confidence LOW on dependent outputs

Unknown	Rule
Income stability	0 rate adjustment, +1 unknown for widening; treated as the more conservative "variable" for the safe-capacity retention table specifically
Repayment history	0 rate adjustment, +1 unknown for widening; no negative safe-capacity or decision-tree penalty
Collateral value	LTV-supported amount not computed; secured lender-likely falls back to FOIR-only, confidence LOW
Emergency savings	No bonus, no penalty (treated as genuinely unknown, not "none")

11. Final Adaptive Question Flow

START

↓

Must questions (9, Stage 1/2 §D — unchanged, no issues raised)

↓

Determine incomeType → branch to Salaried / Self-employed / Informal adaptive set (Stage 2 §4)

↓

Ask branch-specific adaptive questions

↓

Material-uncertainty check (new, formalizing Issues 2 & 3):

- household_expenses still unknown? → ask ONE coarse-bucket follow-up
- high_cost_debt > 0 AND monthly payment unknown? → ask directly (already a required

adaptive question); if still unknown, apply the 25%-of-outstanding fallback

↓

Calculate all outputs, with confidence set per-output per Section 16-methodology below

Question	Trigger	Reason	Output affected	Required before calc?	If unknown
(9 must questions)	Always	Sufficient alone for all 4 outputs	All	Yes (amount/income/type only)	Section 10 defaults
employment_tenure	Salaried	Rate/lender adjustment	Fair rate, Lender	No	0 adjustment, +widening
income_stability	Salaried	Safe-capacity exclusion	Safe amount	No	Conservative "variable" default
variable_income_share	Salaried, if variable	Sizes exclusion	Safe amount	No	Skip, no exclusion applied
repayment_history	Salaried/Self-employed	Rate + decision signal	Fair rate, Verdict	No	0 adjustment, +widening
business_tenure	Self-employed	Lender/rate	Lender, Fair rate	No	0 adjustment, +widening
documented_income (ITR)	Self-employed	Income split	Lender, Fair rate	No	Treated as fully undocumented
income_stability_biz	Self-employed	Retention/stress	Safe amount, Stress	No	Conservative "unstable" default

Question	Trigger	Reason	Output affected	Required before calc?	If unknown
collateral_available	Self-employed (always); Salaried (large ask)	LTV, routing	Lender, Product route, Fair rate	No	Unsecured routing only
existing_business_debt	Self-employed	Competes for cash flow	Lender/ Safe amount	No	Assumed ₹0 (disclosed)
co_applicant	Self-employed/Informal	Joint capacity	Lender, Safe amount	No	Not assumed
productive_return	Self-employed, purpose=business	Framing only (Issue 4)	Explanation text only	No	No effect
income_range	Informal, no point figure	Volatility-aware base	All affordability	Yes if no point income given	N/A — one of the two forms is always required
income_volatility	Informal	Stress parameter	Stress	No	Default –20% used
household_dependency	Informal (always), light (Self-employed)	Disposable income	Safe amount	No	0 adjustment
high_cost_debt	Informal (always), conditional (others)	Hard-stop trigger	Verdict, Safe amount	No	Assumed none

Question	Trigger	Reason	Output affected	Required before calc?	If unknown
high_cost_debt_monthly_payment	high_cost_debt > 0	The burden, not the balance, drives the math	Safe amount, Verdict, Stress	Yes, once high_cost_debt > 0	25%-of-outstanding fallback (Issue 2)
recent_bounce	Informal (always), conditional (others)	Near-term red flag	Verdict, Fair rate	No	Assumed false (disclosed)
emergency_savings_months	All branches	Safe-capacity margin	Safe amount, Stress	No	No adjustment either way
upcoming_large_expense	Large ask relative to income	Pre-fund a known shock	Safe amount	No	No adjustment
household-expense coarse bucket (<i>new, conditional</i>)	household_expenses left blank	Cheap precision gain (Issue 3)	Safe amount	No	Default range shown instead

12. Final Question → Output Impact Matrix

(Unchanged from Stage 2 Section 5 for all surviving questions — no question was cut this round; the impact ratings there remain valid. productive_return remains rated Low/None across all hard-number columns, by design, per Issue 4's resolution, and is retained anyway per that decision.)

13. Corrected Priya Calculation

Input	Value	Type
Age, tenure, income ₹1,10,000, car EMI ₹14,000, rent ₹28,000, credit score 780	as given	FACT (Lokta)
Other essential expenses ≈ ₹20,000/month	—	ASSUMPTION (Lokta doesn't itemize beyond rent; a real user would answer this directly as a USER ANSWER)
No missed payment, income stability = fixed	—	USER ANSWER (implied by "salaried 5 years," treated as a direct answer)

Income: documented = claimed = eligibleIncomeLender = eligibleIncomeSafe = **₹1,10,000** (fully verifiable; income-split model collapses to the trivial case). *Derived.*

Lender capacity: FOIR 50% → maxDebtService ₹55,000 – existing EMI ₹14,000 = availableEMI **₹41,000**. Fair-rate band (below) midpoint = 10.75% → factor(10.75%,36mo)=30.65 → FOIR-supported = 41,000×30.65 ≈ **₹12,56,650**. No collateral offered → not LTV-bound. **Lender-likely ≈ ₹12.57L.**

Safe capacity: disposable = 110,000 – 48,000 (essentials) – 14,000 (EMI) = **₹48,000**. Retention: base 50% (salaried, stable, no adjustments trigger) = **50%**. Safe EMI ceiling = 48,000×0.50 = **₹24,000**. Recommended EMI = 24,000×0.90 = **₹21,600**.

Fair rate (Section 6 algorithm): creditScoreAdj(780,≥750)=-20; repaymentHistoryAdj(clean)=-5; stabilityAdj(stable)=-10; documentationAdj(full)=-5; highCostDebtAdj=0. Σ=-40. finalPosition=clamp(50-40,0,100)=**10**. Unknowns=0 → halfWidth=15. Band positions [0,25] (clamped from [-5,25]). Base band (bank tier, personal loan)=[10,16]. **Fair rate = 10.0%–11.5%.**

Cost: ceiling rate 11.5% used for safe/recommended conversion → factor(11.5%,36mo)=30.32. Safe ceiling principal = 24,000×30.32 ≈ **₹7,27,680**. Recommended principal = 21,600×30.32 ≈ **₹6,54,912**. Effective annualized cost at recommended amount, 2% fee ≈ **13.7%** (worked in Section 7).

Requested: ₹8,00,000 — above safe ceiling (₹7.28L), well within lender capacity (₹12.57L) → **BORROW LESS**, recommend ≈ ₹6.55–7.28L.

Stress (income -20%): ratio moves from Comfortable toward **Tight** at the recommended EMI.

Confidence (per output): Lender capacity — HIGH (all inputs known). Safe capacity — HIGH. Fair rate — MEDIUM (score known, but exact lender-specific pricing isn't). Verdict — HIGH.

14. Corrected Ravi Calculation

Input	Value	Type
14 years kirana, ITR ₹4.2L/yr, cash ₹40–80k/mo, shop ₹45L unencumbered, wife ₹18,000 teaching, no credit history, purpose = business expansion	as given	FACT (Lokta)
No recent missed payment, business income "steady" (not seasonal)	—	ASSUMPTION (Lokta's text implies stability from 14-year tenure but doesn't state this directly; a real Ravi would answer this as a USER ANSWER)
Household essential expenses	not given	ASSUMPTION — default (65% of income) applied per Section 10, confidence flagged LOW on this basis
Wife is a disclosed co-applicant	implied by "wife earns ₹18,000"	ASSUMPTION that she'd be included as co-applicant; flagged, since Lokta's text states her income but not her willingness to co-apply

Income (Section 3): $\text{documentedIncome} = 4,20,000/12 = \text{₹}35,000$.

$\text{claimedTotalIncome} = \text{midpoint of } 40\text{--}80\text{k (tenure } \geq 3\text{y)} = \text{₹}60,000$.

$\text{undocumentedPortion} = \text{₹}25,000$.

- $\text{eligibleIncomeLender (secured, 40\% haircut)} = 35,000 + 0.40 \times 25,000 = \text{₹}45,000$; + wife's documented ₹18,000 → household **₹63,000**.
- $\text{eligibleIncomeSafe (no haircut)} = \text{₹}60,000$; + wife's ₹18,000 → household **₹78,000**.

Lender capacity (secured LAP): FOIR 60% → $\text{maxDebtService} = 63,000 \times 0.60 = \text{₹}37,800$, no existing debt → $\text{availableEMI} = \text{₹}37,800$. Fair-rate midpoint (below) = 11.7% → $\text{factor}(11.7\%, 84\text{mo}) = 57.16$ → FOIR-supported ≈ **₹21,60,648**. Collateral: $45,00,000 \times 0.80 (\text{haircut}) = 36,00,000$; $\times 0.60 (\text{commercial LTV}) = \text{₹}21,60,000$. **Lender-likely = MIN(21.61L, 21.60L) ≈ ₹21.6L** — effectively tied, marginally LTV-bound.

Safe capacity: disposable = 78,000 – 50,700(65% default essentials) – 0 = **₹27,300**.

Retention: self-employed/ITR/steady base **40%**, no adjustments triggered (no bounce disclosed, no stated instability) = **40%**. Safe EMI ceiling = 27,300×0.40 = **₹10,920**.

Recommended EMI = ₹9,828.

Fair rate: thin-file(+5); repaymentHistory clean(–5); stability stable(–10);

documentation partial(0); highCostDebtAdj(0). $\Sigma = -10$.

finalPosition=clamp(50–10,0,100)=**40**. Unknowns=0 → halfWidth=15 → band [25,55].

Base band (secured commercial LAP) = [9.5,15]. Rate(25)=9.5+0.25×5.5=**10.9%**;

Rate(55)=9.5+0.55×5.5=**12.5%**. **Fair rate = 10.9%–12.5%**. Midpoint used above for lender conversion = 11.7%.

Cost: safe/recommended conversion at ceiling rate 12.5%, 84mo → factor=51.83. Safe ceiling principal = 10,920×51.83 ≈ **₹5,66,000**. Recommended principal = 9,828×51.83 ≈ **₹5,09,400**.

Requested: ₹15,00,000 — comfortably within lender capacity (₹21.6L), well above safe capacity (₹5.1–5.7L) → **BORROW LESS**, recommend ≈₹5.1–5.7L via LAP, with an explicit Card note that this gap is unusually wide and its size is driven mainly by the **defaulted, undisclosed household-expense figure** — the highest-value question to actually ask Ravi before finalizing.

Confidence (per output): Lender capacity — MEDIUM (no formal valuation, income split estimated). Safe capacity — LOW (expenses defaulted). Fair rate — MEDIUM. Verdict — MEDIUM.

15. Corrected Anita Calculation

Input	Value	Type
Income ₹26,000–30,000/mo, two children, husband unemployed 8 months, ₹35,000 outstanding at 30%+ APR, one bounce last month, wants ₹1,50,000 for a scooter	as given	FACT (Lokta)
Monthly payment toward the ₹35,000 debt	not given by Lokta	ASSUMPTION — the app's fallback rule (25% of outstanding, Issue 2) applied because no figure exists; a real Anita would give a USER ANSWER instead of this fallback

Input	Value	Type
Household essential expenses	not given	ASSUMPTION — default (65% of income), confidence LOW
Emergency savings	not given	genuinely UNKNOWN — no adjustment applied either way (not assumed zero)

Income: documentedIncome = ₹0 (no ITR, no payslip). claimedTotalIncome = lower bound of range (gig, high volatility) = ₹26,000. undocumentedPortion = ₹26,000.

- eligibleIncomeLender (unsecured, 10% haircut) = $0 + 0.10 \times 26,000 = ₹2,600$.
- eligibleIncomeSafe = ₹26,000.

High-cost-debt EMI (Issue 1/2, demonstration assumption): outstanding ₹35,000 × 25% = ₹8,750/month — *labeled explicitly as a demonstration assumption applying the app's own fallback rule, not a Lokta-provided fact.*

Lender capacity: FOIR (informal/undocumented) 35% → maxDebtService = $2,600 \times 0.35 = ₹910$. Minus highCostDebtEMI ₹8,750 → **negative, floored at ₹0. Lender-likely ≈ ₹0 (mathematical capacity only).**

Safe capacity: disposable = $26,000 - 16,900(65\% \text{ default essentials}) - 8,750(\text{high-cost debt}) = ₹300$. Retention: informal base 25% – 10(bounce) – 5(dependents >2, non-earning spouse) = **10%** (at floor). Safe EMI = $300 \times 0.10 = ₹30/\text{month}$ (negligible).

Sensitivity check per Issue 3: even at the low-expense end of the default range (50% of income), disposable = 4,250 → safe EMI ≈ ₹425/month — still negligible against a ₹1,50,000 ask. The Don't-Borrow conclusion is robust to the expense uncertainty.

Hard-stop check: burden = $8,750 / 26,000 = 33.65\% \geq 30\% \rightarrow$ **SEVERE hard stop triggers on its own**, independent of the bounce. (The COMPOUND condition — burden $\geq 15\%$ AND bounce — is also independently satisfied, reinforcing the same conclusion via a second path.)

Verdict: DON'T BORROW — driven by the hard stop, not by the (already near-zero) capacity numbers; both paths agree.

Fair rate (reference only, per Issue 5 algorithm): creditScore unknown(0, +1 widening); repaymentHistory bounce(+20); stability unstable(+10); documentation none(+15); highCostDebtAdj(+10). $\Sigma = +55$. finalPosition = clamp(50+55, 0, 100) = **100**. halfWidth = $15 + 10(1 \text{ unknown}) = 25 \rightarrow$ band [75, 100]. Base band (NBFC/weak-tier personal loan) = [16, 30]. Rate(75) = $16 + 0.75 \times 14 = 26.5\%$; Rate(100) = **30%+**. **Fair rate = 26.5%–30%+** — shown as reference only, explicitly not actionable given the verdict.

02/03/04, per Issue 5 (must still appear): lender $\approx ₹0$, safe $\approx ₹30\text{--}425/\text{month}$, fair rate 26.5–30%+, stress = **already Unsustainable at baseline**, before any shock — all four are shown, all labeled "mathematical capacity — not a recommendation to borrow."

Confidence (per output): Lender capacity — LOW (income fully undocumented, EMI defaulted). Safe capacity — LOW (expenses and EMI both defaulted). Fair rate — LOW.

Verdict — HIGH (the two facts that actually drive it — the bounce and the $\geq 30\%$ APR debt — were directly disclosed by Lokta, not assumed).

16. Final Negotiation Cards

Priya

Decision: Borrow Less

Amount: Requested ₹8,00,000 · Likely lender ₹12.57L · Safe ceiling ₹7.28L · Recommended ₹6.55–7.28L

Rate: 10.0%–11.5%

Cost: ~2% processing fee · Estimated effective annualized cost $\approx 13.6\text{--}13.9\%$

EMI: Safe ceiling ₹24,000/mo · Illustrative EMI at recommended amount $\approx ₹21,600/\text{mo}$

Stress (income –20%): Tight

Why: Your existing car EMI and rent already use a meaningful share of your monthly capacity;

the full ₹8L would push past what's comfortable even though a lender would likely approve it.

Confidence: High on affordability, Medium on rate.

Checklist: negotiate the processing fee · confirm fixed vs floating · never exceed ₹24,000/mo.

Ravi

Decision: Borrow Less (via secured route)

Likely lender route: Loan against property (shop premises), not unsecured business/personal loan

Amount: Requested ₹15,00,000 · Likely lender (LTV/FOIR-bound) $\approx ₹21.6\text{L}$ · Safe ceiling $\approx ₹5.66\text{L}$ ·

Recommended $\approx ₹5.1\text{--}5.7\text{L}$

Rate: 10.9%–12.5% (secured) vs. 16%+ if taken unsecured

Cost: ~1% processing fee (typical for LAP) · Estimated effective annualized cost ≈11–13.5%

EMI: Safe ceiling ≈₹10,920/mo · Illustrative EMI at recommended amount ≈₹9,828/mo

Why: Your documented (ITR) income alone would struggle to support ₹15L unsecured, but your

unencumbered shop supports much more via LAP than is actually safe for your household's

month-to-month cash flow — the biggest driver of that gap is an estimate, not a real number,

for your household expenses.

Confidence: Medium on lender capacity (no formal valuation), Low on safe capacity (expenses

estimated) — answering the expense question would sharpen this significantly.

Checklist: get a formal property valuation before finalizing · confirm residential vs

commercial LTV classification · ask whether your wife's income can be included as co-applicant.

Anita

Decision: Don't Borrow

Amounts shown below are mathematical capacity only — NOT a recommendation to borrow:

Lender capacity ≈ ₹0 · Safe capacity ≈ ₹30–425/month · Fair rate (reference only) 26.5%–30%+

Why: You already pay an estimated ₹8,750/month toward existing debt at 30%+ interest — over

30% of your income — and a payment bounced last month. A new ₹1,50,000 loan would add cost on

top of debt you haven't cleared. This conclusion holds even if your actual expenses turn out to

be lower than we assumed.

What to consider instead: look at consolidating or renegotiating the existing high-cost debt

before any new borrowing; revisit the scooter once that debt and the missed payment are resolved.

Confidence: High on the decision itself (the facts driving it were directly disclosed);

Low on the exact rupee figures (your monthly debt payment and household expenses were estimated,

not confirmed — please provide these for a sharper picture).

17. Final Assumption Register

(Consolidating Section 2's audit table with category tags — every number below appears exactly once as canonical.)

Category	Count	Examples
Assignment-derived	2	Income-shock magnitude (−20%), rate-shock magnitude (+2 pts) — both explicitly suggested in the Lokta brief
External fact	7	Gold-loan LTV tiers (RBI 2026), LAP residential LTV (70%), LAP commercial LTV (60–65%, conservative end used), personal-loan bank/NBFC bands, gold-loan bands, home-loan bands, processing-fee typical range
Product assumption	~10	FOIR percentages, collateral valuation haircut, unknown-EMI fallback (25%), household-expense defaults, existing-EMI unknown floor, tenure defaults, documentation haircuts
My judgement	~25	Retention factors and all their adjustments, presentation headroom factor (90%), all fair-rate position adjustments, stress classification thresholds, high-cost-debt thresholds (30%/15%), request/safe ratio trigger, confidence-band counting rule

(Full line-by-line register is Section 2's table — this is a categorical summary as requested, not a duplicate.)

18. Final Engineering Contract

INPUT LAYER

QuestionnaireAnswers { all must + adaptive fields, raw as entered }

PROFILE LAYER (normalization — pure function of QuestionnaireAnswers)

BorrowerProfile { incomeType, documentedIncome, claimedTotalIncome, undocumentedPortion, eligibleIncomeLender, eligibleIncomeSafe, existingEMI, businessDebtEMI, highCostDebtEMI (+ isDefaulted flag), essentialExpenses (+ isDefaulted flag), collateral{type, statedValue}, creditScore/status, repaymentHistory, stability, documentationStatus, dependents, emergencySavingsMonths, recentBounce, requestedAmount, purpose, productPreference }

CALCULATION LAYER (pure functions, no UI dependency)

lenderCapacity(profile) → { fairSupportedAmount, ltvSupportedAmount, bindingConstraint,

amount, confidence, explanation }

safeCapacity(profile) → { disposableCashFlow, retentionFactor, adjustments[], safeEMI, recommendedEMI, safeAmount, recommendedAmount, confidence, explanation }

fairRate(profile) → { baseBand, startingPosition, adjustments[], finalPosition, halfWidth, lowerRate, upperRate, confidence }

effectiveCost(principal, rate, fee, tenure) → { emi, netProceeds, effectiveMonthlyRate, effectiveAnnualizedCost, includedItems[], excludedItems[] }

stressTest(profile, safeCapacity, fairRate) → { incomeShockResult, rateShockResult, classification, confidence }

RULES LAYER (decision logic — depends only on calculation-layer outputs, never re-touches raw

inputs directly)

decision(profile, lenderCapacity, safeCapacity, stressTest) → { verdict, hardStopsTriggered[],

softSignalsTriggered[], reason, confidence, drivers[] }

productRoute(profile) → { route, requiresCollateral, tradeoffs[] }

confidence(...) → per-output HIGH/MEDIUM/LOW via the counting rule
(Section 16

of Stage 2, unchanged) — never a single blended number

OUTPUT LAYER

DecisionResult, LenderCapacity, SafeCapacity, FairRate, Cost, StressResult —
structured objects

as specified in Stage 2 Section 24, unchanged, each carrying value + confidence +
explanation

+ drivers[]

PRESENTATION LAYER

NegotiationCard (assembles all of the above; adds the fixed checklist; applies the
"mathematical capacity, not a recommendation" label whenever verdict = DON'T
BORROW)

Hard constraints for Stage 3: the calculation layer must never receive
productive_return as an argument (Issue 4's enforcement mechanism);
presentationHeadroomFactor (0.90) is applied in exactly one place (safeCapacity(),
when deriving recommendedAmount/recommendedEMI) and nowhere else in the
codebase; no network call, ML model, or LLM may sit inside the calculation or rules
layers.

19. Model Limitations

- The retention-factor table, the 30%/15% high-cost-debt thresholds, and the fair-rate adjustment magnitudes remain **my judgement**, not externally verified — defensible by direction (safe < lender, compound signals escalate, single signals don't) more than by the exact numbers.
- Business-loan and two-wheeler rate bands were never independently re-verified this session and remain the weakest-sourced bands in the model.

- Ravi's and Anita's safe-capacity numbers both rest on **defaulted household-expense figures** the real personas were never asked for — the single highest-value piece of missing data across all three walkthroughs, and the Card says so in both cases.
- The 25%-of-outstanding high-cost-debt fallback is a deliberately conservative single point drawn from a wider plausible range (15–30%); a real borrower's actual repayment structure could sit outside even that range.
- No bureau pull, no bank-statement parsing, no property valuation — every number is self-reported and unverified by construction; this is disclosed, not hidden.
- A real lender's manual underwriting discretion, relationship pricing, and lender-specific risk appetite are not modeled; this is a market-level estimate, not a specific-lender prediction.

20. FINAL RULES FREEZE

These are the frozen rules for Stage 3.

Domain	Canonical formula/threshold
Income	$\text{eligibleIncomeLender} = \text{documentedIncome} + \text{haircut}(10\%/40\%) \times \text{undocumentedPortion}$; $\text{eligibleIncomeSafe} = \text{claimedTotalIncome}$ (no haircut)
Lender capacity	Lender-likely = FOIR-supported (unsecured) or MIN(FOIR-supported, LTV-supported) (secured); FOIR = 50/45/35/60% by segment; LTV = 85/80/75% (gold, tiered), 70% (LAP residential), 60% (LAP commercial); collateral haircut 20%
Safe capacity	$\text{safeEMI} = \text{disposable} \times \text{retentionFactor}(\text{clamped } 10\text{--}55\%)$; $\text{recommendedEMI} = \text{safeEMI} \times 0.90$ (applied once, universal, purpose-independent)
Fair rate	Position model: $\text{start } 50 + \sum \text{adjustments}$, clamp 0–100; $\text{halfWidth} = 15 + 10 \times \text{unknownCount}$; linear mapping onto [lowerRate, upperRate]
Cost	$\text{netProceeds} = \text{EMI} \times [1 - (1+r')^{-n}] / r'$, solved for r' , annualized $(1+r')^{12} - 1$; labeled "Estimated Effective Annualized Borrowing Cost"
Stress	Income –20% / rate +2pts; classify $\leq 35/36\text{--}45/46\text{--}55/>55$ as Comfortable/Tight/Stressed/Unsustainable

Domain	Canonical formula/threshold
Decision	Hard stops: burden $\geq 30\%$ (severe) OR (burden $\geq 15\%$ AND bounce) OR (bounce AND baseline Unsustainable); else compare requested vs. safe vs. lender; soft signals (≥ 2) escalate one notch
High-cost debt	APR $\geq 30\%$ defines it; unknown monthly payment = outstanding $\times 25\%$
Confidence	Per-output HIGH/MEDIUM/LOW via missing-input count, never a single blended score

Two engineers implementing this document independently will produce the same lender/safe/rate/cost/stress/verdict numbers for Priya, Ravi and Anita, within rounding — verified above. Every number that isn't an external, sourced fact is explicitly tagged **product assumption / my judgement** and is defensible on that basis in a live follow-up.

Stage 3 may now begin implementation exactly as specified.